

SHINE STANDARD INTEGRATED SCIENCE

PUPIL'S BOOK 5

FIRST EDITION



For Excellence in Integrated Science

PEREZ.K

BASED ON THE CURRENT CURRICULUM



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BY

AP SHINE STANDARD PUBLISHERS

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We do sincerely regret any mistakes which may be found anywhere in this book. It is not intended to be part of this book but accidental.

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Author: Kirimuttu Perez

Typesetting: Kirimuttu Perez

Cover design: Shine Standard Graphics

Illustrations: Shine Standard Graphics

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PREFACE

Shine Standard Integrated Science Pupils' Book Five has been developed basing on the revised **Primary Five Integrated Science Syllabus** as prescribed by the new curriculum of the **National Curriculum Development Center (NCDC)**. The book contains accurate, relevant and current information covering all topics in all terms of the year in their order. It is intended to guide both teachers and learners.

The pupil's book cares for the interests of the learners in terms of simple and concise language used, simplified content to cater for all learners with different abilities and clear illustrations to make learning enjoyable through observation. Key words for each topic have also been included in order to enrich the learner's vocabulary and mastery of concepts.

The topics have well organized, relevant, and easy to understand notes and facts. It is written in a simple language and is well aided with illustrations/diagrams where necessary to ease understanding.

The book is remarkably precise but detailed in content with **no fact left hanging**. It has been mainly written for Primary Five in a language that is suitable for both rural and urban Pupils. The book can therefore be used with minimum teacher guidance.

The book has inbuilt and continuous assessment activities. These questions are to help the learners to test their understanding of the concepts covered and are to enable the teacher to track progress as coverage goes on. This also makes the book convenient for individual and class learning by the candidates.

The activities are set basing on the three main levels of assessment at primary level that is to say: **Knowledge, Comprehension and Application**. The questions are carefully set in order to challenge the pupils in thinking, recalling, observing, comprehending and applying knowledge and skills in day-to-day life. These tested skills are also key to excellent performance at PLE since it tests all the above levels.

Projects and experiments have been included to make the subject practical and to enable learners discover knowledge and facts by themselves. This is also to promote a positive attitude towards Integrated science as a subject.

The book is written and developed by experienced teachers of Integrated Science and we welcome all comments on the publication with an open mind for the improvement in the teaching and learning of Integrated Science. Comments and orders can be communicated **directly through the following contacts:**

TEL: +256 705 104620, ☎ +256 774 339588 ☎ +256 775 809996

Email address: shinestandard22@gmail.

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THEME: SCIENCE IN HUMAN ACTIVITIES AND OCCUPATIONS

TOPIC 1: KEEPING POULTRY AND BEES

Poultry keeping

-Poultry keeping is the rearing and management of domestic birds or fowls.

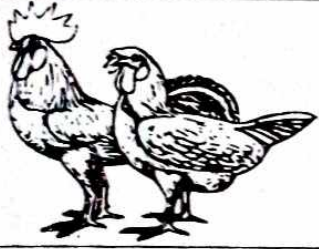
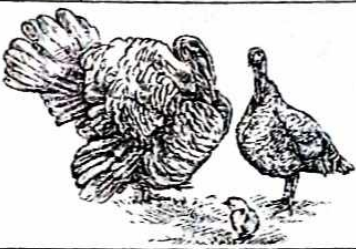
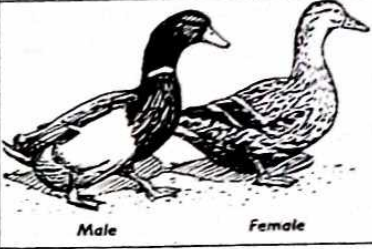
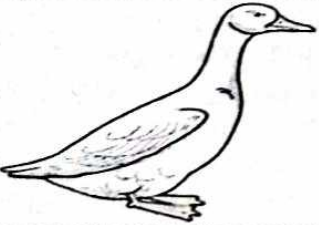
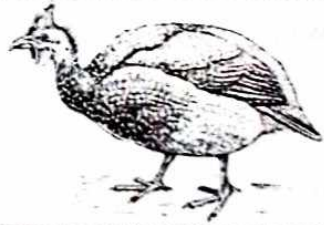
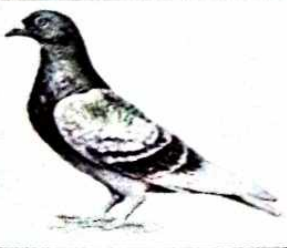
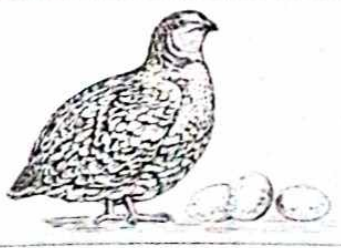
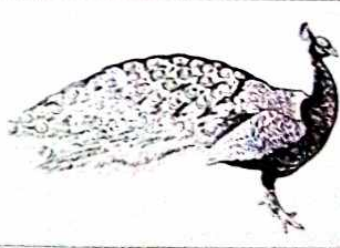
Poultry

-Poultry refers to all kinds of domestic birds.

-Domestic birds can also be called fowls.

Examples of poultry

-Chickens (domestic fowls), -Turkeys, -Ducks, -Geese, -Doves, -Guinea fowls, -Pigeons, -Quails, -Peacocks, Parrots etc.

Chickens	Turkeys	Ducks
		
-These are the commonest bird among poultry. -They are kept for meat, eggs and cultural purposes.	-These are the biggest among poultry. -Kept for meat and eggs -Their bare skin on the neck changes colour from grey to red when excited or unhappy.	-These have webbed feet for swimming and spoon shaped beak for getting food from water. -Kept for meat, eggs, and sometimes feathers.
A goose	A dove	Guinea fowl
		
-These resemble ducks though longer than ducks. -They have webbed feet for swimming.	-These are commonly kept by people as pets.	-These are originally wild birds, though some are domesticated. -They provide meat and eggs.
A pigeon	A quail	A peacock
		
-These are also kept as pets -Some breeds are raised for their meat.	-They provide nutritious eggs and meat.	-They are kept as pets and for meat in some cultures.

Note: The following are the common breeds of:

i) Ducks

-Khaki Campbells - Indian Runners

ii) Turkeys

-American Bronze - Mammoth White Turkeys

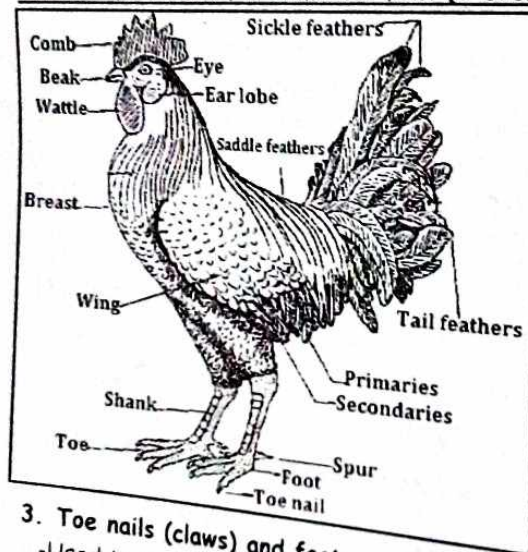
Reasons for keeping poultry/Uses of poultry

- Some birds are kept for egg production.
- Some birds are kept for meat production.
- For study purposes.
- The cocks help to tell time when they crow.
- Poultry droppings are used as farm yard manure.
- Some birds are kept as pets. E.g. peacocks and parrots.
- People keep poultry for cultural purposes like dowry, rituals and sacrifices.
- Poultry keeping is a source of income and employment to people.
- To get feathers for different uses e.g. making pillows, decoration, dancing props etc.
- The bones of some birds can be used in the making of buttons, glue and fertilizers.

The domestic fowl

-This refers to chicken kept at home.

External features of a domestic fowl (a cock)



Importance of each part on a bird

1. Spur - It is used for protection / defence

2. Beak / bill

- Used picking food.
- Used for turning eggs by the hens.
- It is where the bird's nostrils are located.
- It is used for defence and fighting.
- Used for building their nests.
- It supports the bird during mating.
- Used to clean itself and arrange its feathers (Preen)

3. Toe nails (claws) and foot

- Used to scratch the ground for food.
- They are used for fighting and defence.
- Used for walking.
- Used for climbing.

4. Comb and wattle

- Used for identification.
- Wattles help a bird to cool its body on hot days.
- Wattles are also used for attracting mates.

5. The eyes - these are used by birds to see.

6. Nostril - it is used for smelling and breathing.

7. Wings - these are used by birds to fly.

8. Earlobe - protects the ears which are used for hearing.

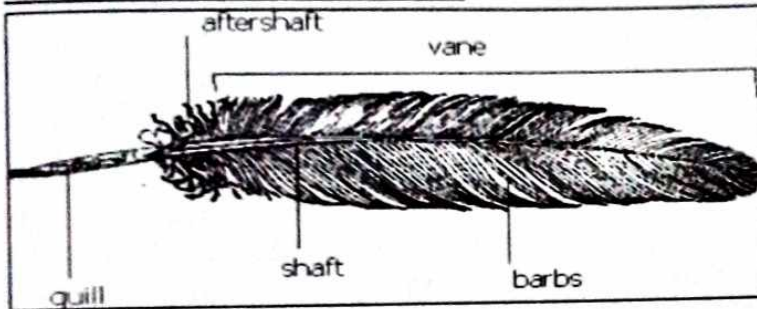
9. The tail - it helps the bird to change the direction during its flight.

10. Feathers - These have different uses as stated below:

Uses of feathers to birds

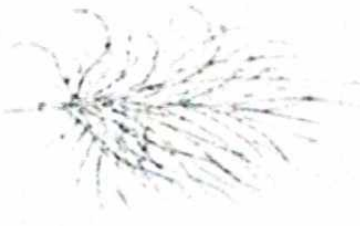
- Feathers keep the bird's body warm.
- Feathers are used for brooding young ones.
- Feathers provide shape to the bird.
- Feathers protect inner body parts of a bird from mechanical injuries.
- The feathers give the birds colour for identification.
- For courtship (attracting of opposite sex for mating) in case of a cock.

Note: Cocks have bright feathers for courtship (Attracting of opposite sex for mating)

Parts of a feather (Quill feather)**Types of feathers**

- These are mainly:

- i) Quill feathers ii) Covert feathers iii) Down feathers iv) Hair (Filoplume) feathers.

1. The Quill feather (Flight feather) - Quill feathers are found on the tail and wings. - They are primary feathers. - They are used for flying (flight)	
2. The body /Covert feathers/contour feathers - They are secondary feathers. - They cover the body of the bird. - They give the bird's body shape. - They keep the bird's body warm.	
3. The down feathers - These are the feathers the chick is hatched with. - In old birds, they lie beneath the body feathers. - They help in insulating the bird's body. - They keep the bird warm.	
4. Filoplume/hair feather - These are the smallest feathers found nearest to the skin of the bird. - They can be seen clearly after removing the body and down feathers	